Nguyen Khanh

Curriculum Vitae

July 2026

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Education

2022 ⋮ 2026 | **Vietnam National University HCMC- University of Information Technology**
B.Sc in Computer Science
GPA: 8.23/10

Projects

2024 | **Home Credit - Credit Risk Model Stability Competition (Kaggle)**

- **Engineered** a highly stable credit risk classification model optimized to resist data drift and maintain predictive performance over temporal shifts.
- **Architected** a robust ensemble pipeline integrating ***LightGBM***, ***XGBoost***, and ***CatBoost***, utilizing ***Time-Series Cross-Validation*** to prevent future data leakage.
- **Optimized** model stability by implementing ***Adversarial Validation*** for feature selection and integrating a custom Gini Stability metric into the early stopping mechanism.

Course Recommendation System (MOOCCubeX Dataset)

- **Architected** a hybrid personalized course recommendation system utilizing the ***MOOCCubeX dataset***, integrating traditional collaborative filtering with deep learning-based recommenders.
- **Implemented and evaluated** a diverse portfolio of recommendation architectures, including ***BPRMF***, ***CBF***, ***NFM***, and ***KGAT*** to capture complex *user-item-concept* interactions.

2025 | **Hybrid Document Retrieval System & Evaluation Pipeline (BEIR Benchmark)**

- **Developed** a hybrid document retrieval system combining semantic dense vectors and lexical sparse features ***(BM25+)***.
- **Architected** a robust NLP preprocessing pipeline to evaluate multiple term-selection strategies (*Lemmatization*, *POS Tagging*, *NER*, *Porter-Stemmed N-grams*) and **established** an automated evaluation workflow on the ***BEIR benchmark.***
- **Engineered** a custom ***Claude MCP*** server to expose the hybrid retriever as dynamic tools for LLM agents.

U²-Net Optima: Resource-Efficient Gastrointestinal Tumor Segmentation (Kvasir-SEG)

- **Architected** an optimized nested U-structure segmentation framework for medical lesion delineation, achieving an ***81.65% mIoU*** and ***87.59% Dice Score*** on the benchmark ***Kvasir-SEG dataset.***
- **Designed** a hardware-efficient training pipeline utilizing ***PyTorch AMP*** and ***Gradient Accumulation*** to minimize VRAM footprint.

2026 | **F5-ASM SIEM-AI: Automated Incident Response & MLOps Pipeline**

- **Architected** a secure, automated SOC-as-code pipeline leveraging ***Grafana Loki*** and ***Promtail*** to ingest, label, and normalize millions of raw ***F5-ASM log events*** in real-time.
- **Engineered** an asynchronous processing engine to aggregate discrete logs into ***IP-based contextual behavior*** sequences, optimize data payload before model inference and map unstructured threat behaviors against the global ***MITRE ATT&CK*** framework.
- **Deployed** an interactive, human-in-the-loop incident response mechanism via *Telegram Bot API.*

Skills

Knowledge	NLP, LLM, Information Retrieval, Computer Vision, Recommender Systems, MLOps
Frameworks & Libs	PyTorch, Scikit-learn, FastAPI, LangChain, LightGBM, XGBoost, CatBoost
Programming	Python, C++, SQL
MLOps & Tools	Docker, AWS, Grafana Loki, Promtail, Model Context Protocol (MCP), Git, Shell, LaTeX
English	IELTS 6.0

Awards and Honors

2023 | ***GPA 4.0 Incentive Scholarship***
2024 | ***Silver Medal (Ranked 163/3856) - Home Credit Kaggle Competition***